

4(a)(i)	vibrations (of particles) are parallel to direction of energy propagation	B1
4(a)(ii)	waves meet/overlap (at a point)	B1
	(resultant) displacement is sum of individual displacements	B1
4(b)(i)	$\lambda = ax / D$	C1
	$= (3.7 \times 10^{-4} \times 4.3 \times 10^{-3}) / 2.3$	C1
	$= 6.9 \times 10^{-7} \text{ (m)}$	A1
	$= 690 \text{ nm}$	
4(b)(ii)	- no change to fringe separation/fringe width/number of fringes - bright fringes are darker - dark fringes are brighter <i>Any two marking points, 1 mark each</i>	B2